MOST IMPORTANT TOPICS FOR GATE CSE

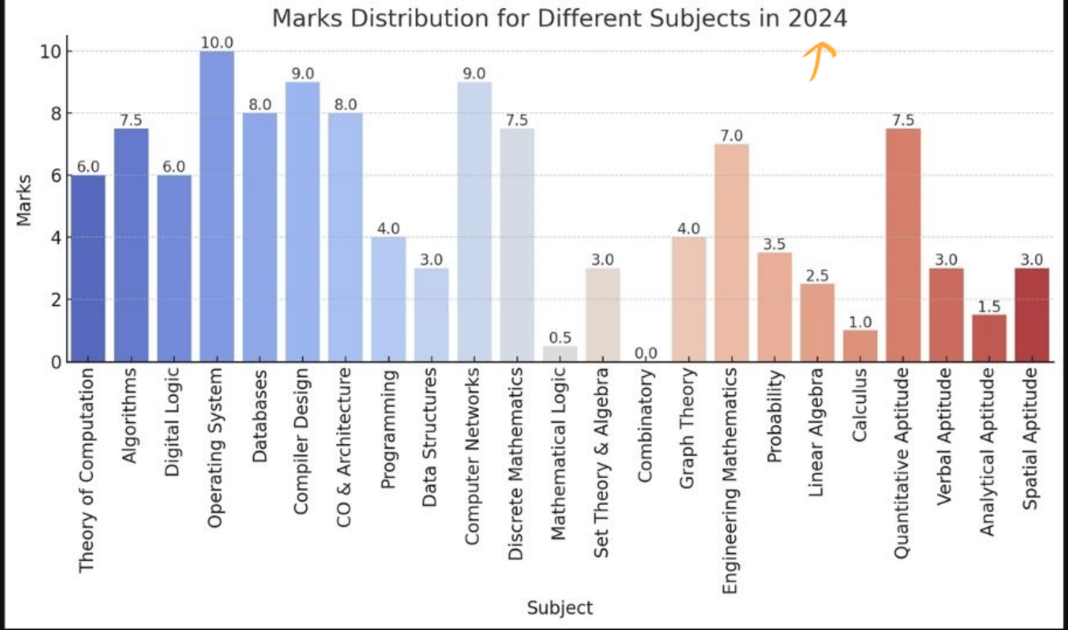
Subjectwise Analysis

Anjali Chauhan

AIR 13 GATE CSE 2023

Last 15 year analysis

https://gateoverflow.in/blog/15807/subjectwise-mark-distribution-in-gate-cse



Marks Distribution for Different Subjects in 2024
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Subject
Theory of Computation 6.0
Algorithms 7.5
Digital Logic 6.0
Operating System 10.0
Databases 8.0
Compiler Design 9.0
CO & Architecture 8.0
Programming 4.0
Data Structures 3.0
Computer Networks 9.0
Discrete Mathematics 7.5
Mathematical Logic 0.5
Set Theory & Algebra 3.0
Combinatory 0.0
Graph Theory 4.0
Engineering Mathematics 7.0
Probability 3.5
Linear Algebra 2.5
Calculus 1.0
Quantitative Aptitude 7.5
Verbal Aptitude 3.0
Analytical Aptitude 1.5
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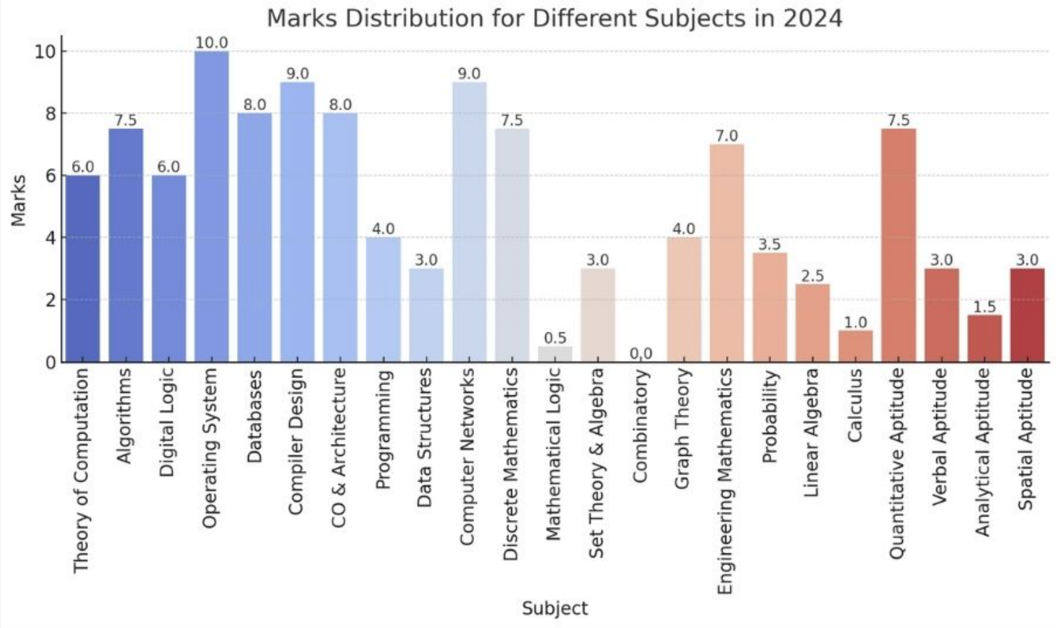
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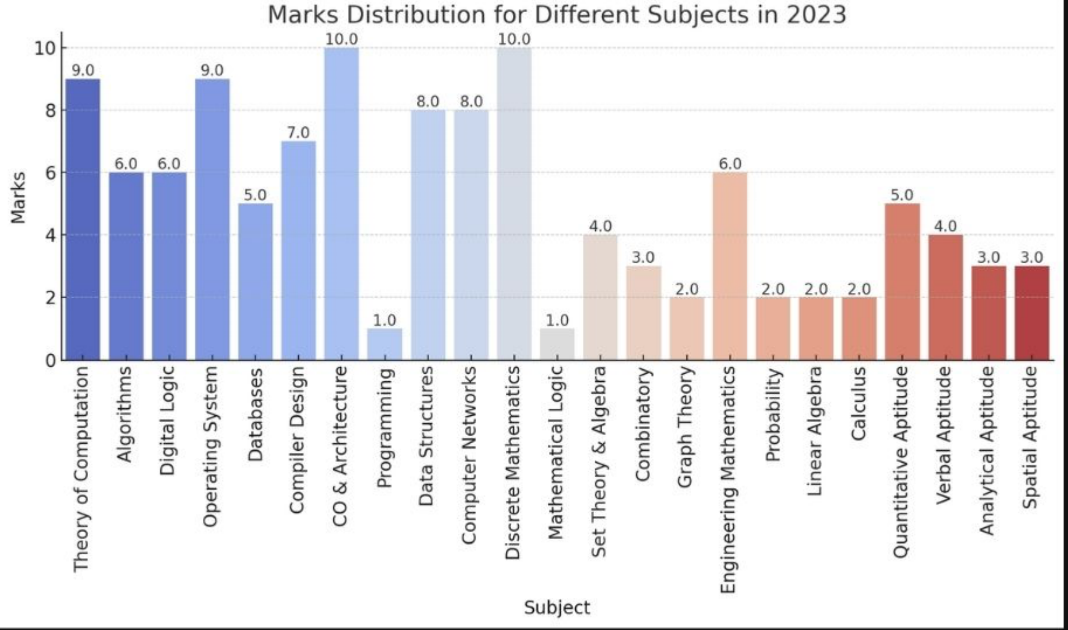
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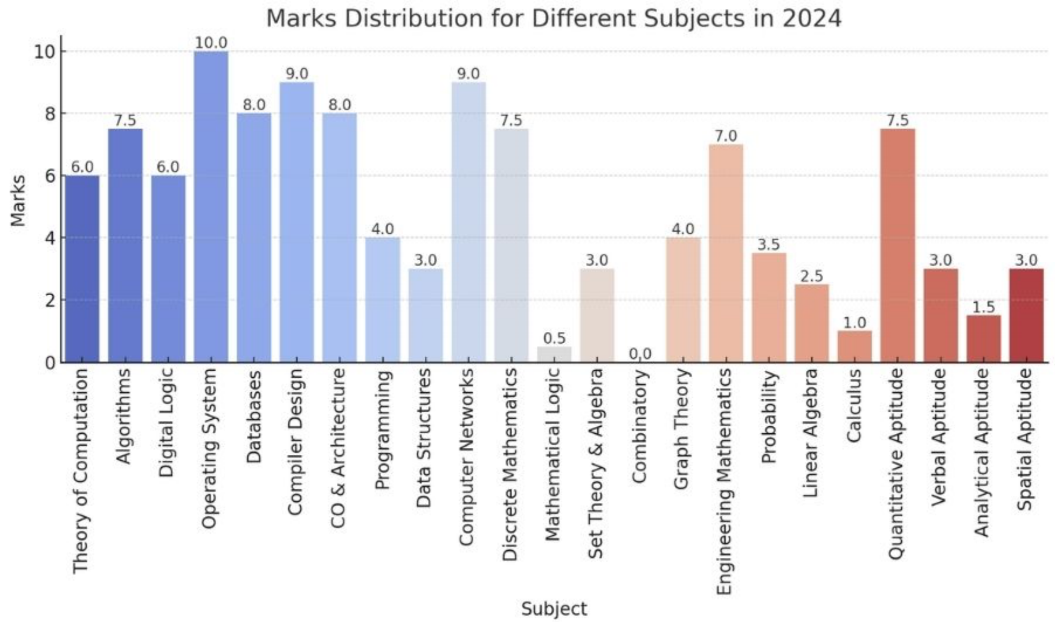
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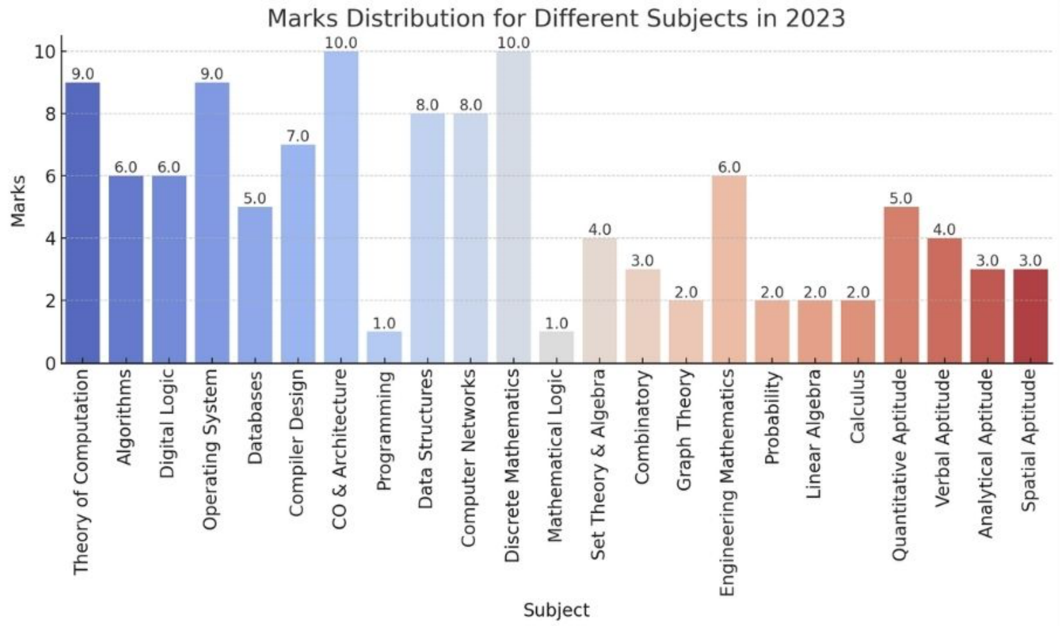
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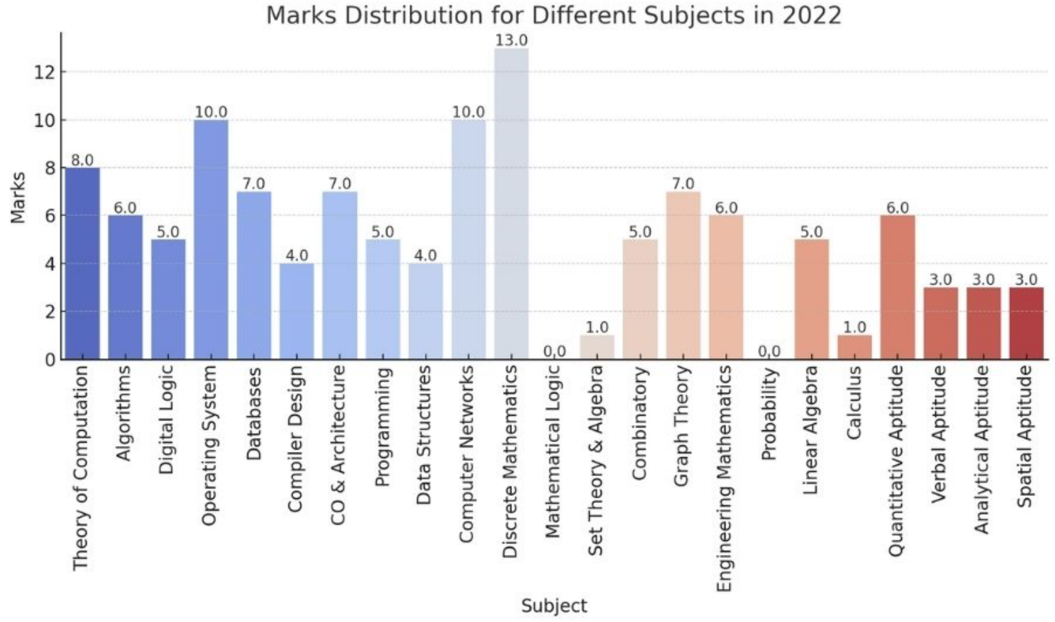
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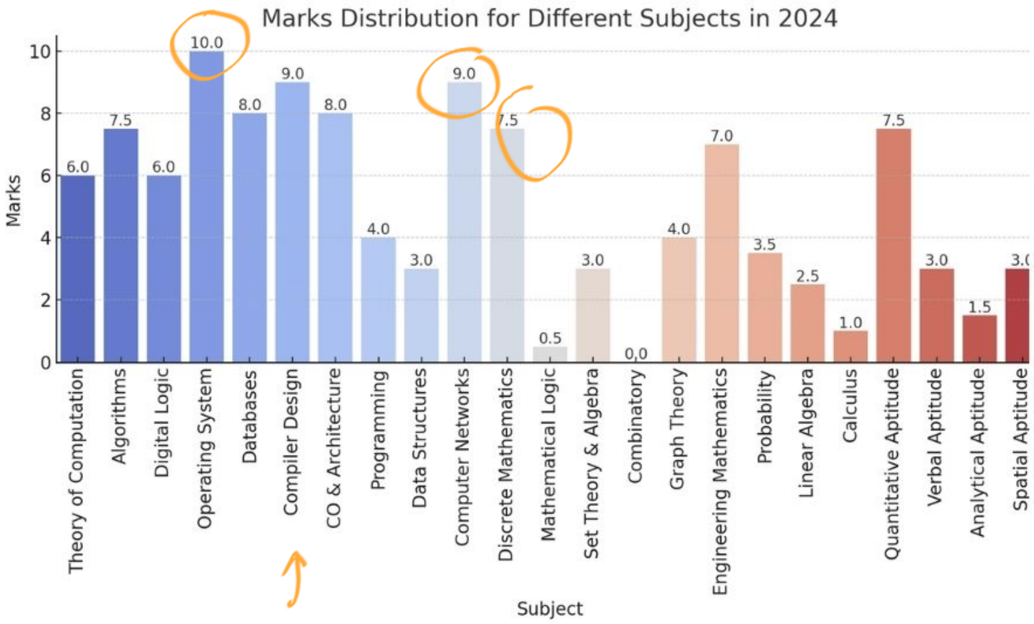
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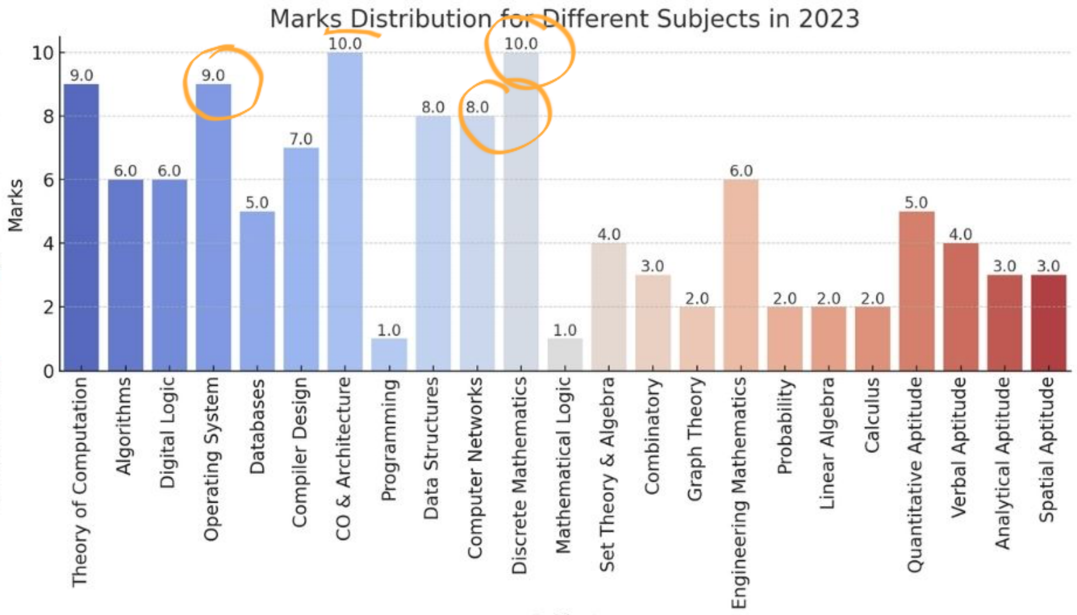
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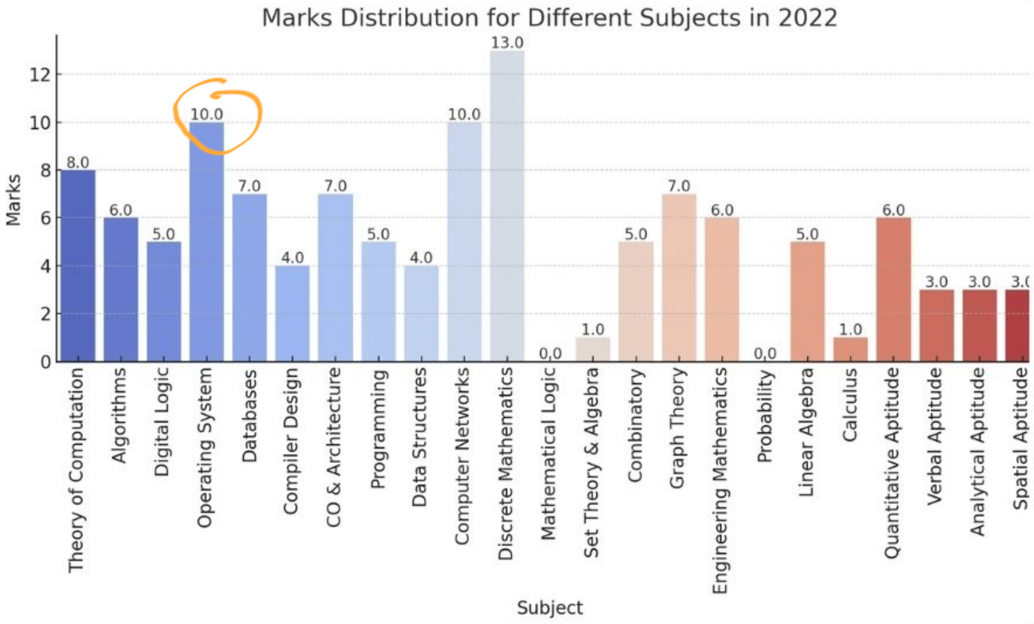
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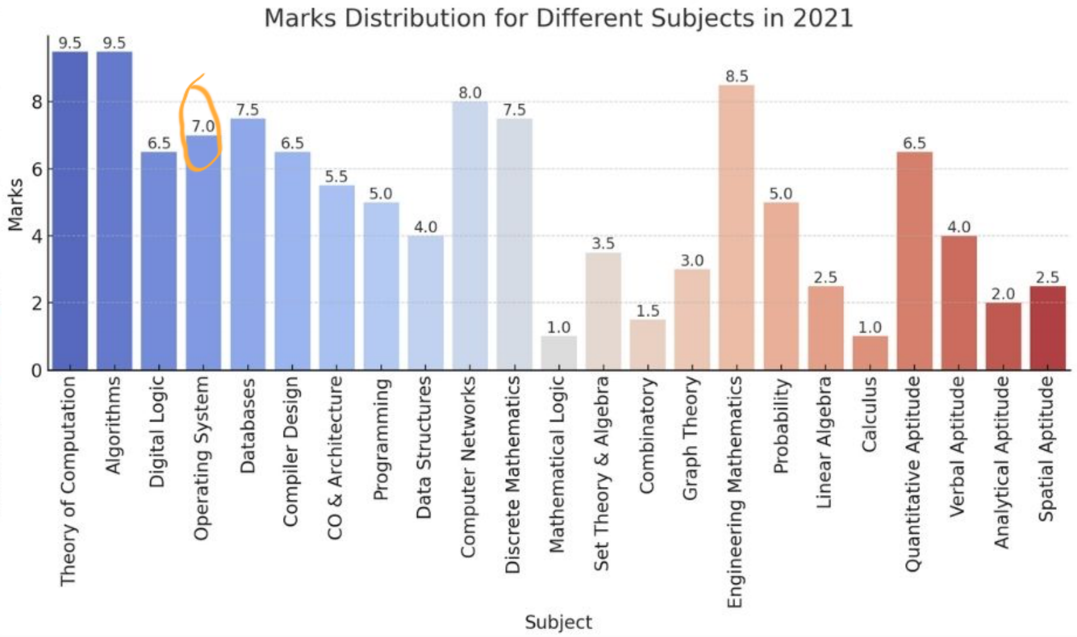
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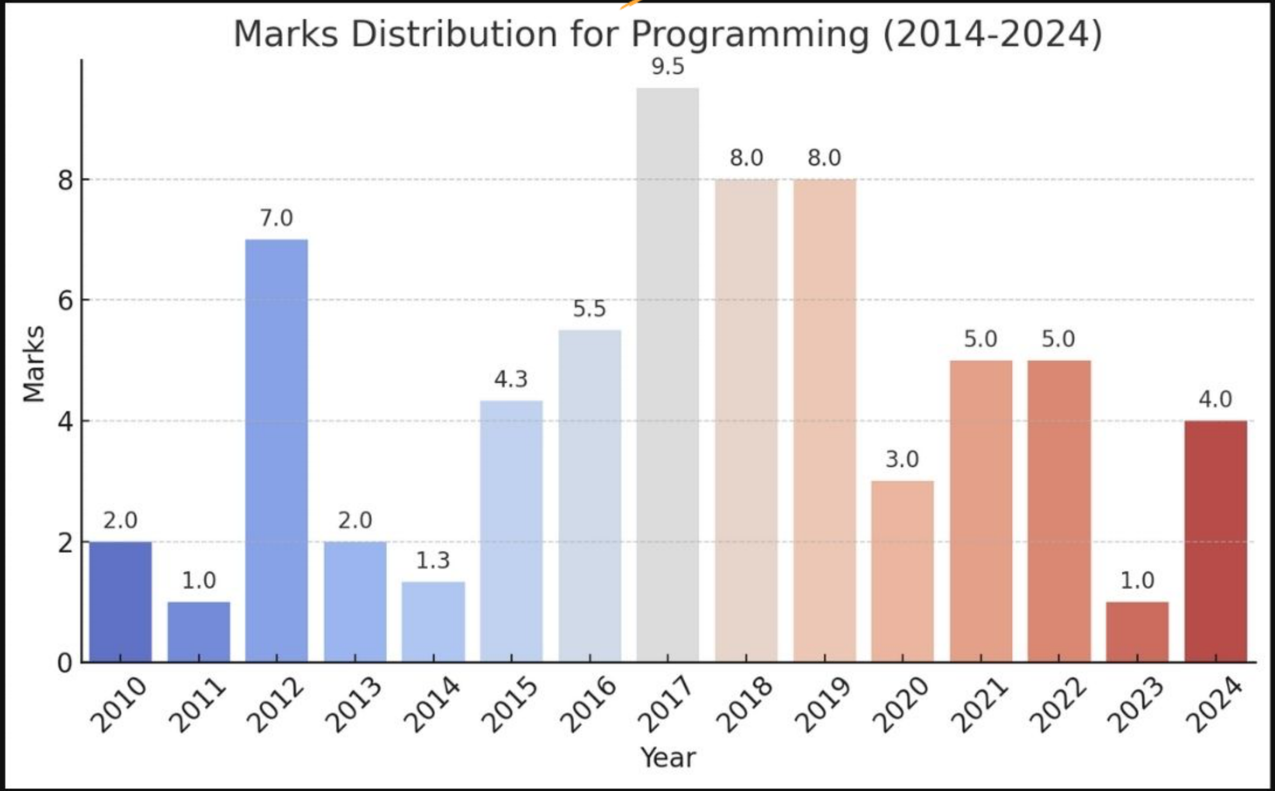
Programming & Data Structures



Marks Distribution for Programming (2014-2024)
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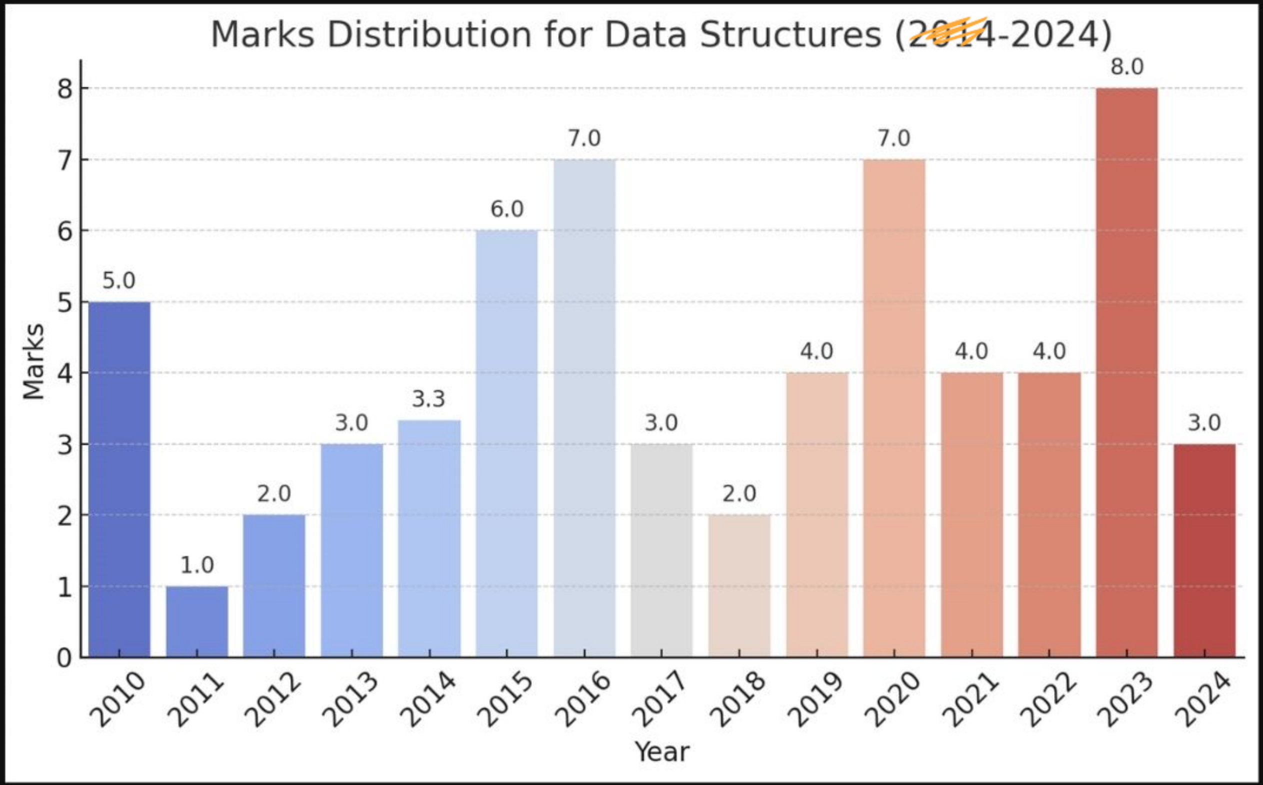
2010-2024



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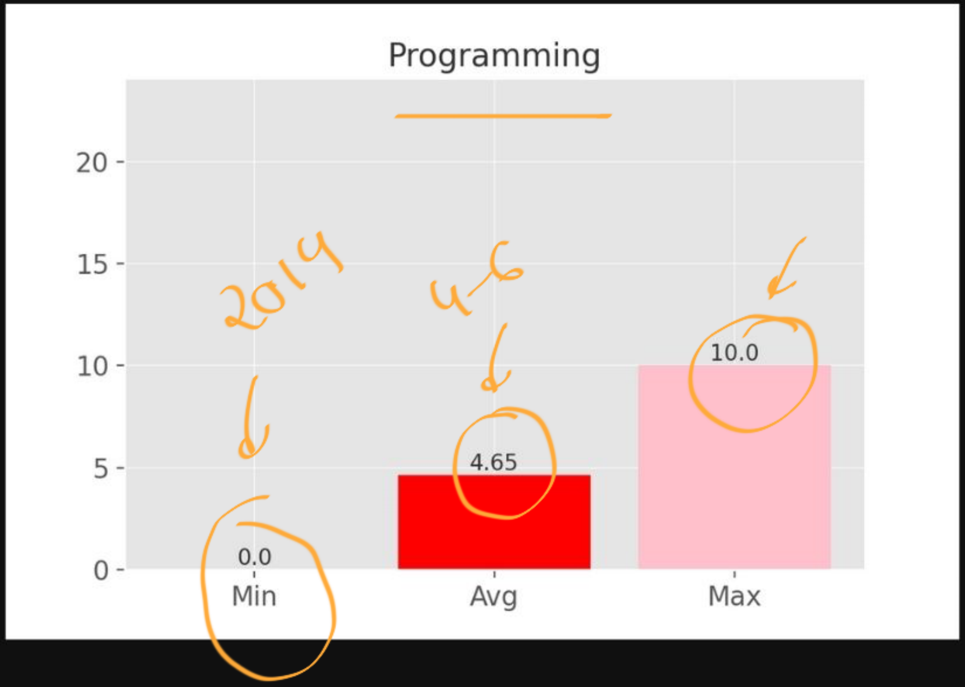
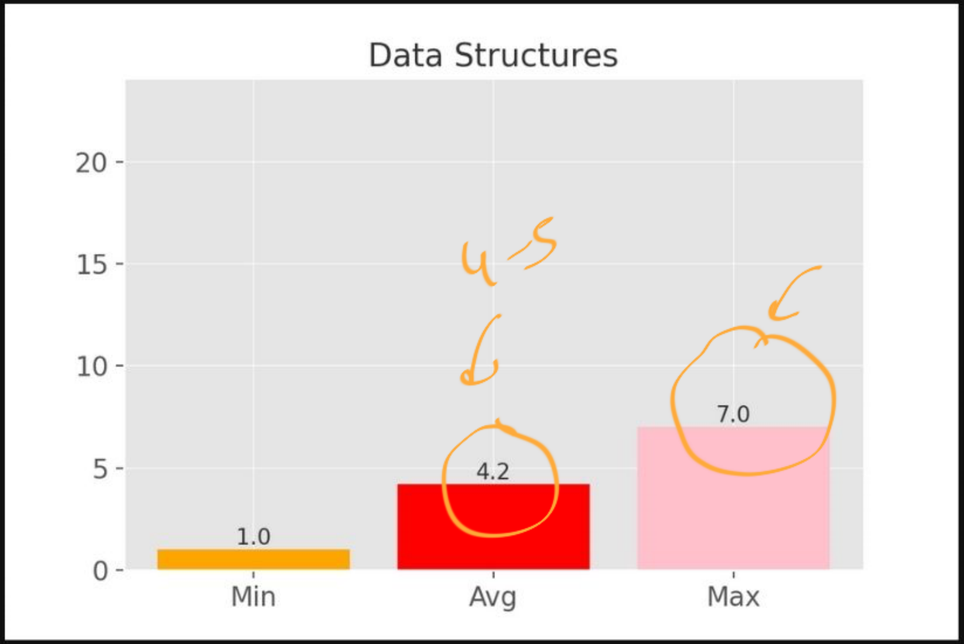
Programming & Data Structures

Min Max Avg marks of last 15 years



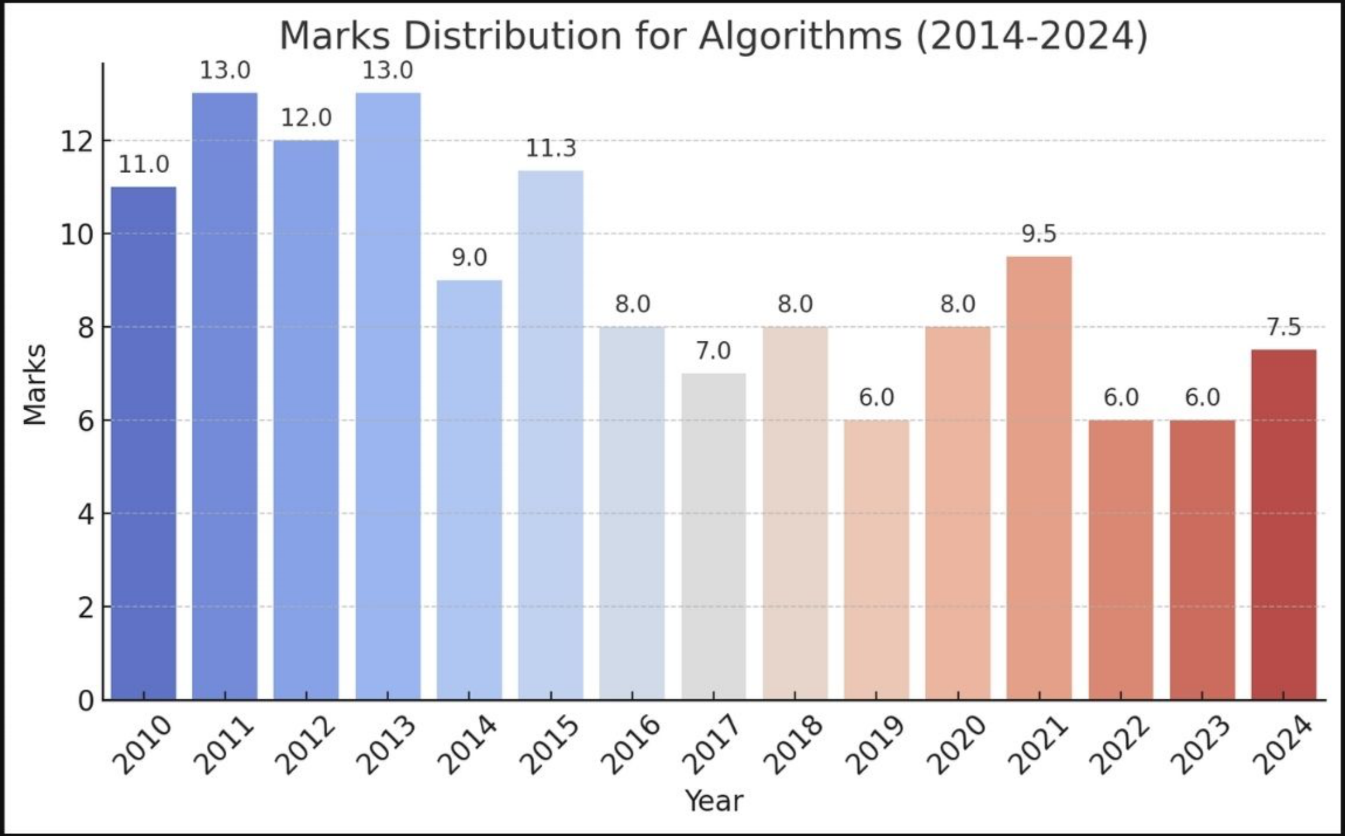


Important Topics

Programming & Data Structures

- Stacks and Queues (29) → easy
- Binary Trees, Binary Search Trees (87) → huge
- Heaps (29)
- Linked List (22)
- Hashing (16) → Tricky
- Arrays (13)

Algorithms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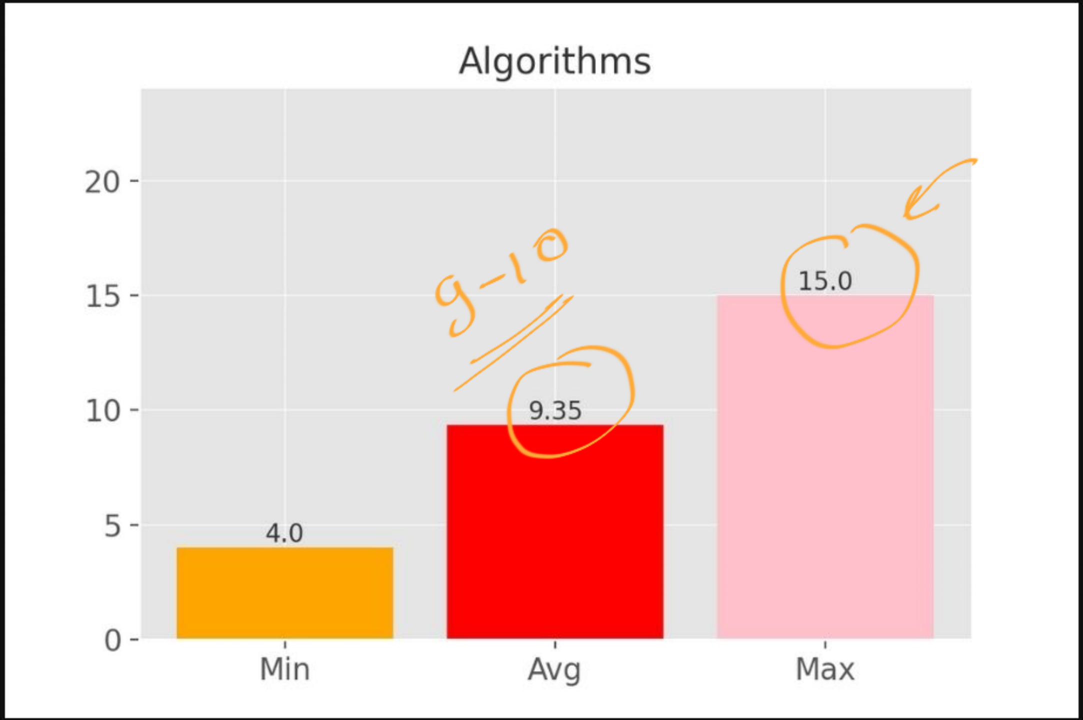
Algorithms

Min Max Avg marks of last 15 years



Important Topics

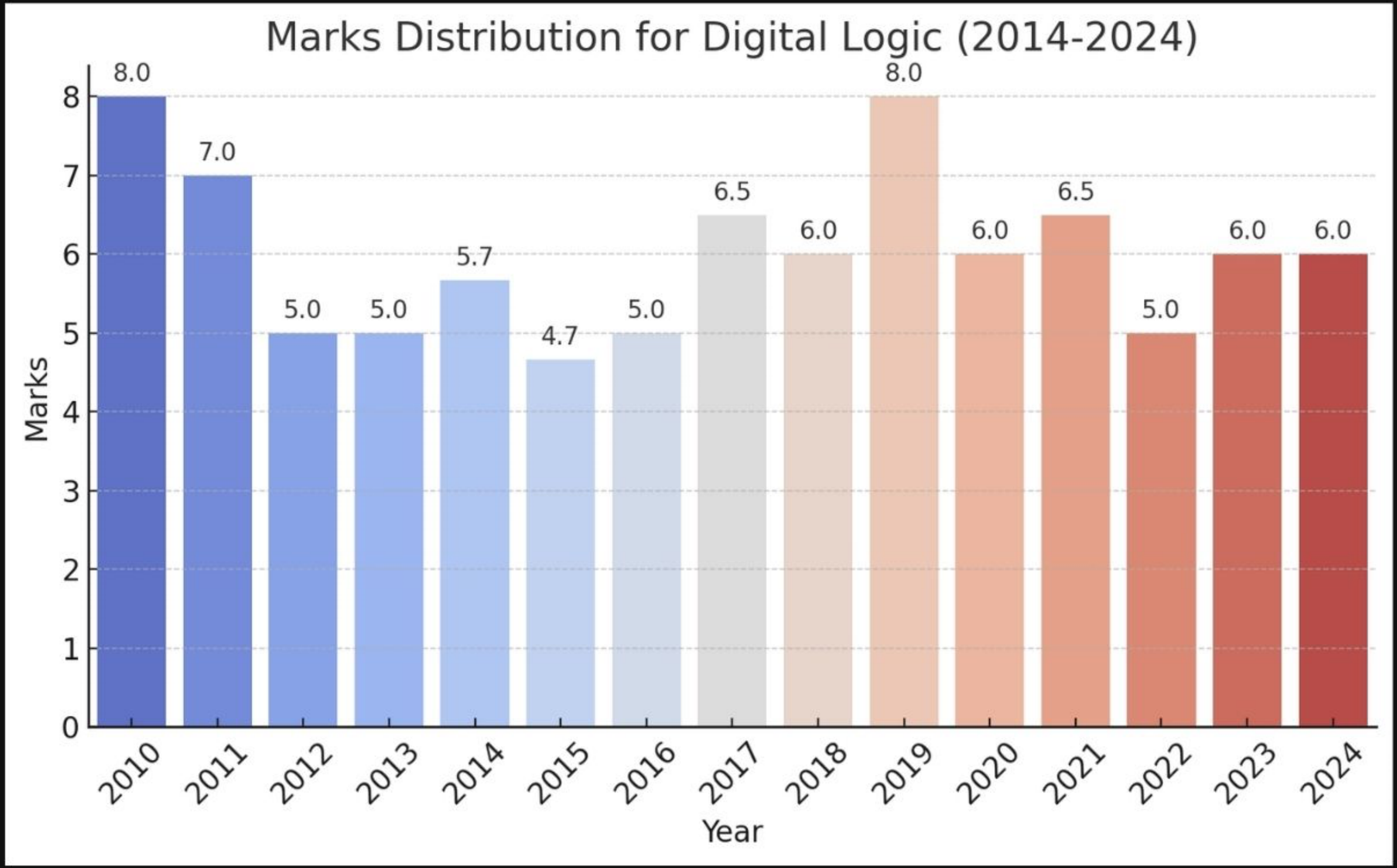
Algorithms

- Asymptotic Notations (21)
- Graph Search (22) → BFS, DFS
- Minimum Spanning Trees (33) → Prims, Kruskal's
- Reccurence Relation (34)
- Sorting (27)
- Time Complexity (29)

easy

easy

Digital Logic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Avg
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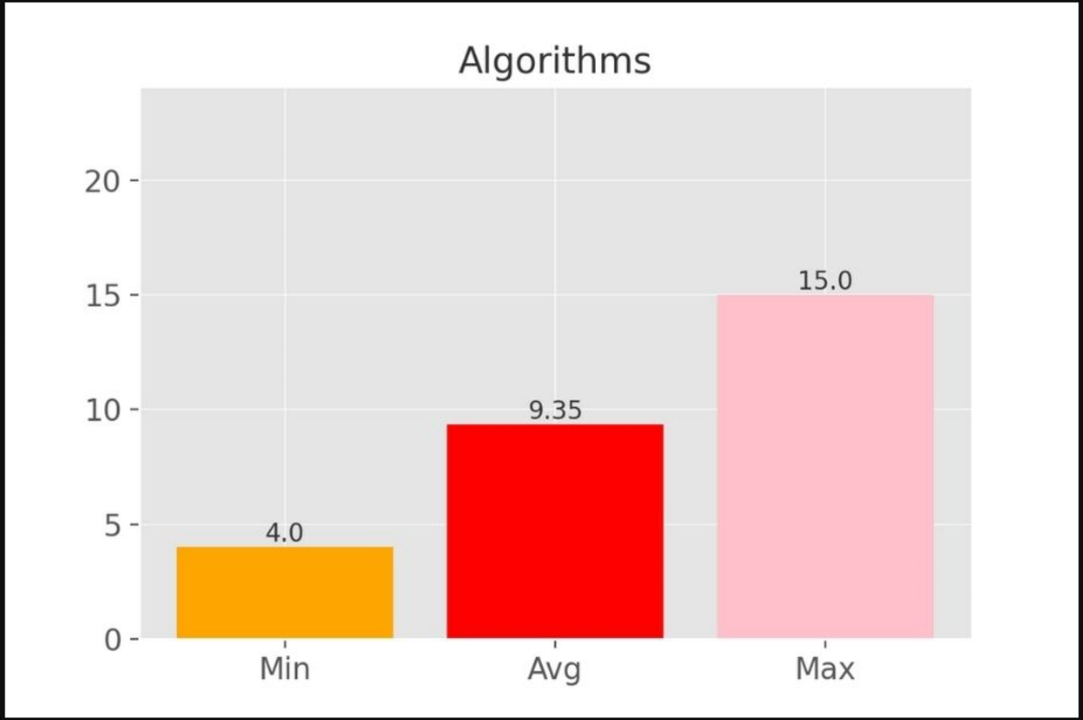
Digital Logic

Min Max Avg marks of last 15 years



Important Topics

Digital Logic

easy

- Boolean Algebra (32)
- Circuit Output (39)
- Number Systems (57)

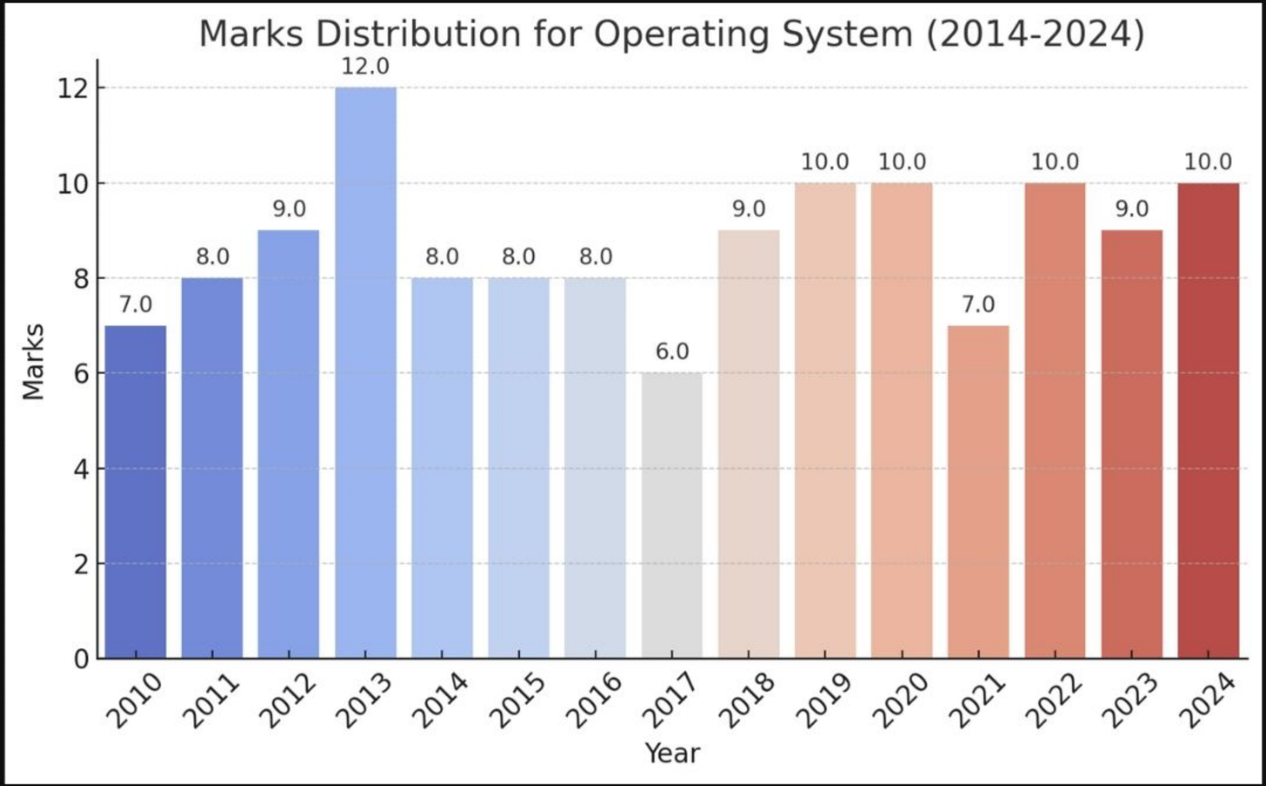
easy

Combinational

Sequential

Tricky

Operating System



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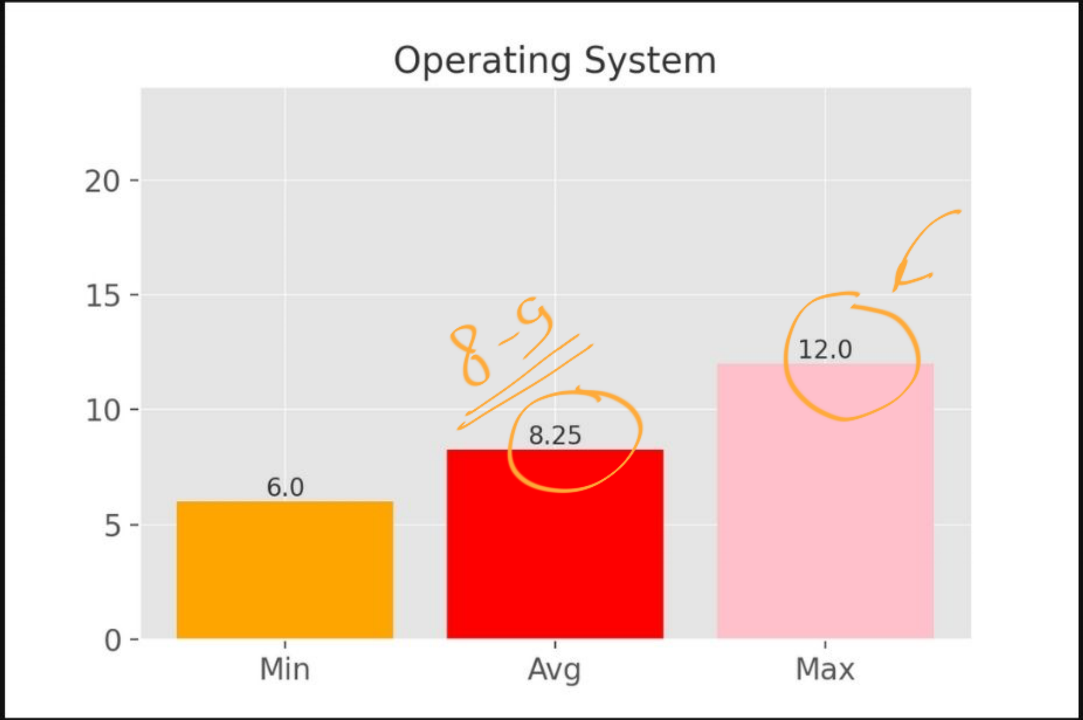
Operating System

Min Max Avg marks of last 15 years



Important Topics

Operating System

easy

- Process Scheduling (47)
- Page replacement (32)

Tricky

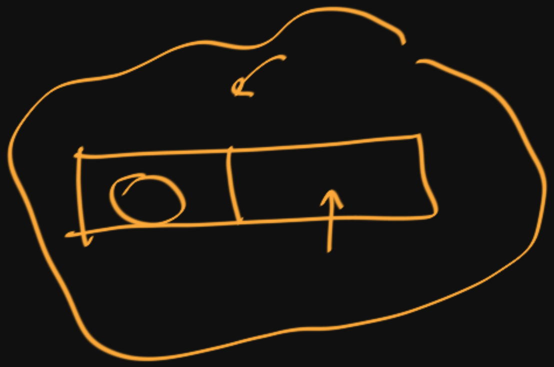
- Synchronization (52)

easy-mod.

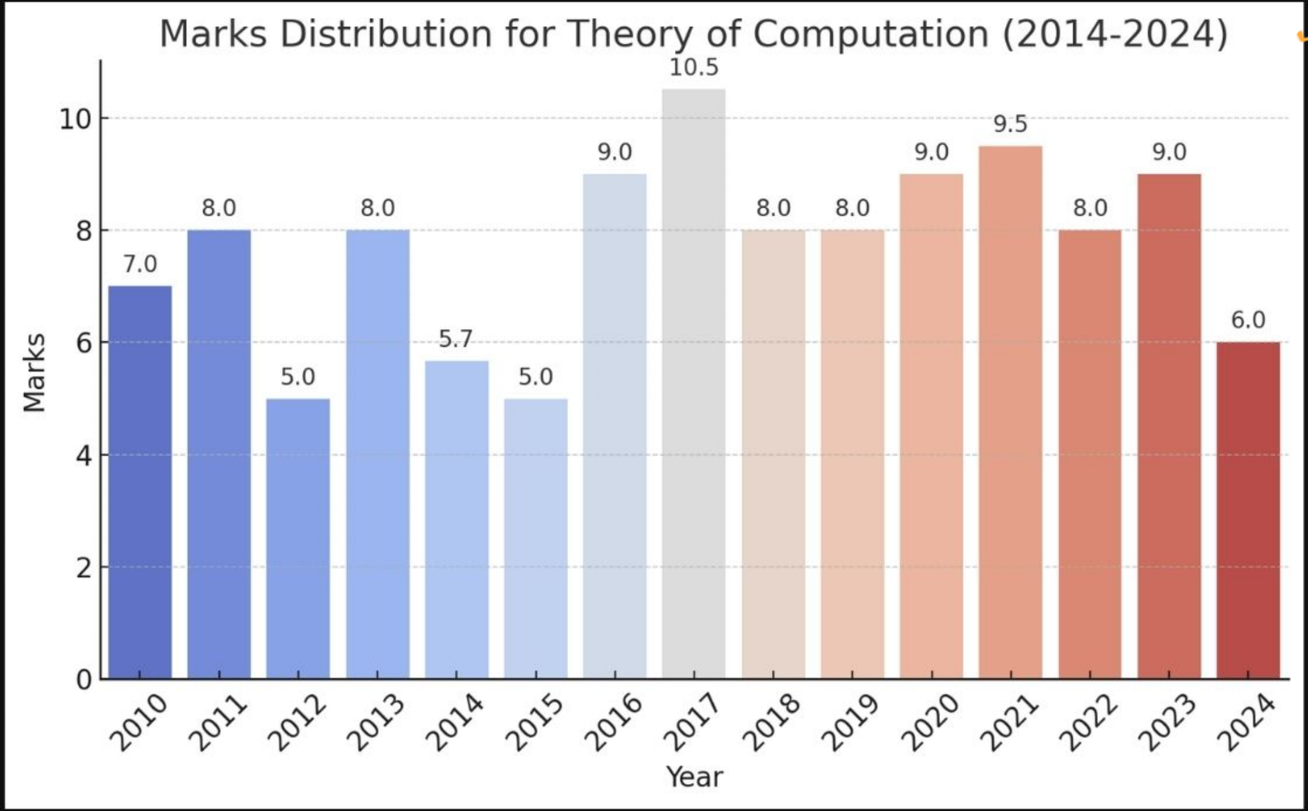
- Disk (45)
- Resource Allocation (27)

Tricky

- Virtual Memory (43)



Theory of Computation



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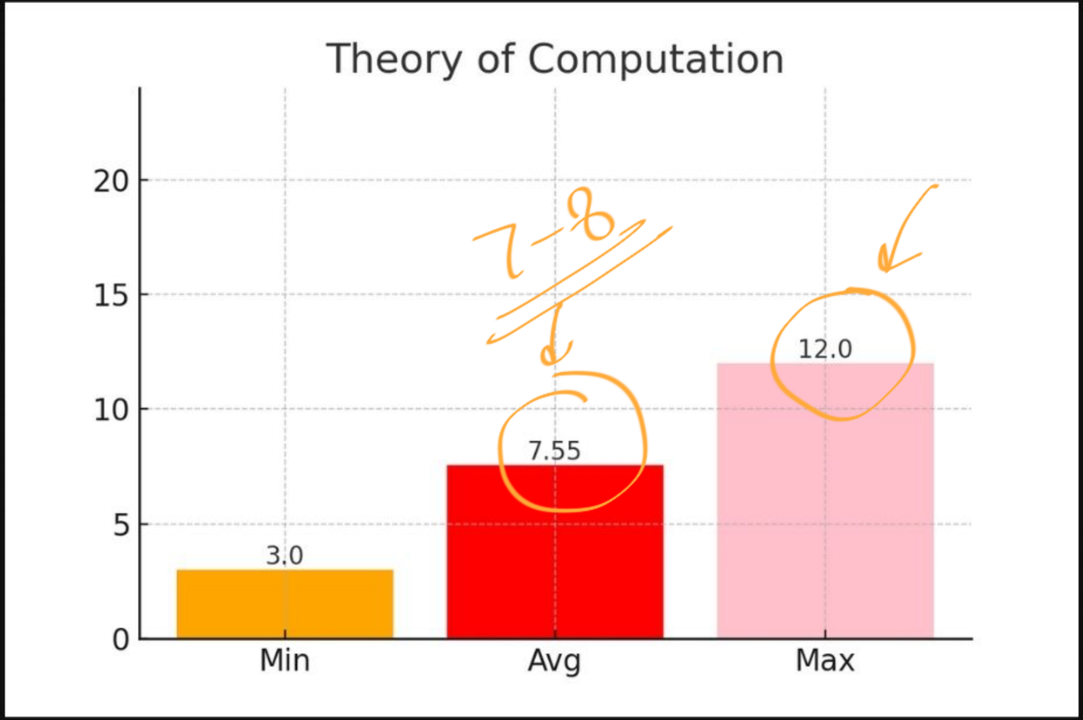
Theory of Computation

Min Max Avg marks of last 15 years

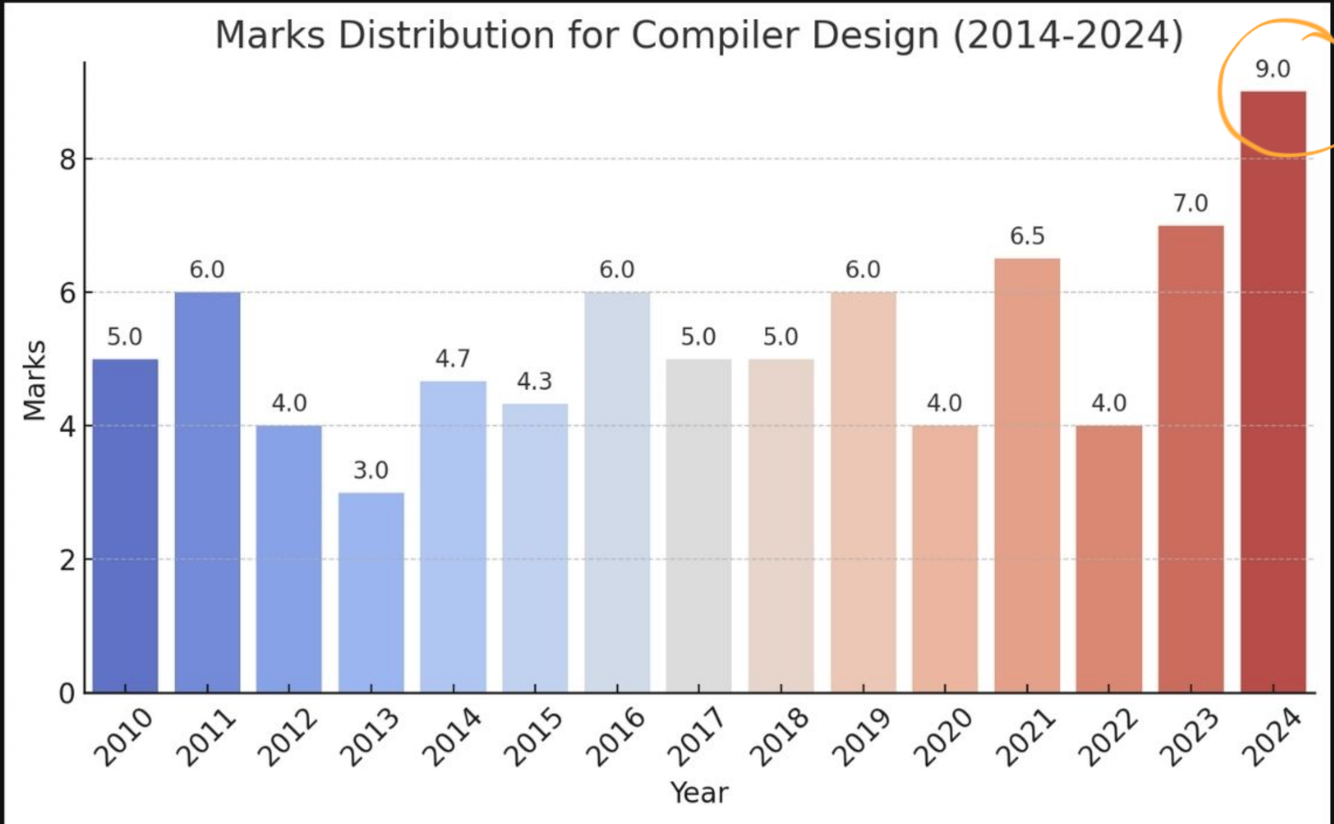


Important Topics

Theory of Computation

- easy Regular Languages, Regular Expressions (62)
- easy-mod Finite Automata (41)
- Context Free Language (32) ← Tricky
- Decidability (29) ← Tricky

Compiler Design



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set 1
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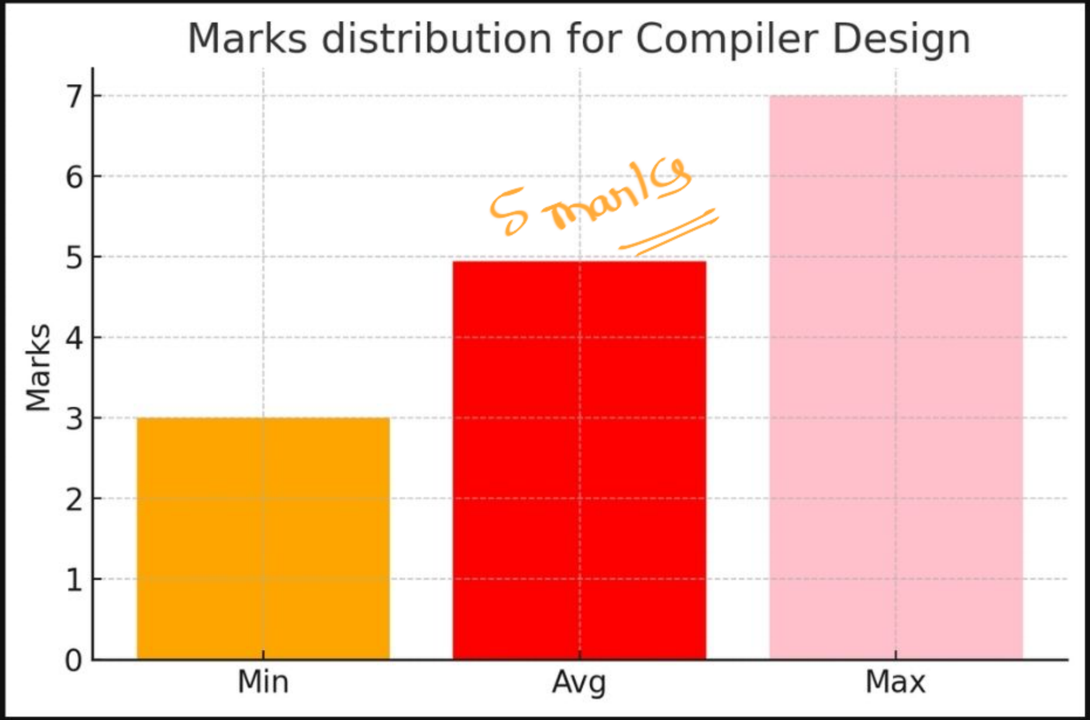
Compiler Design

Min Max Avg marks of last 15 years

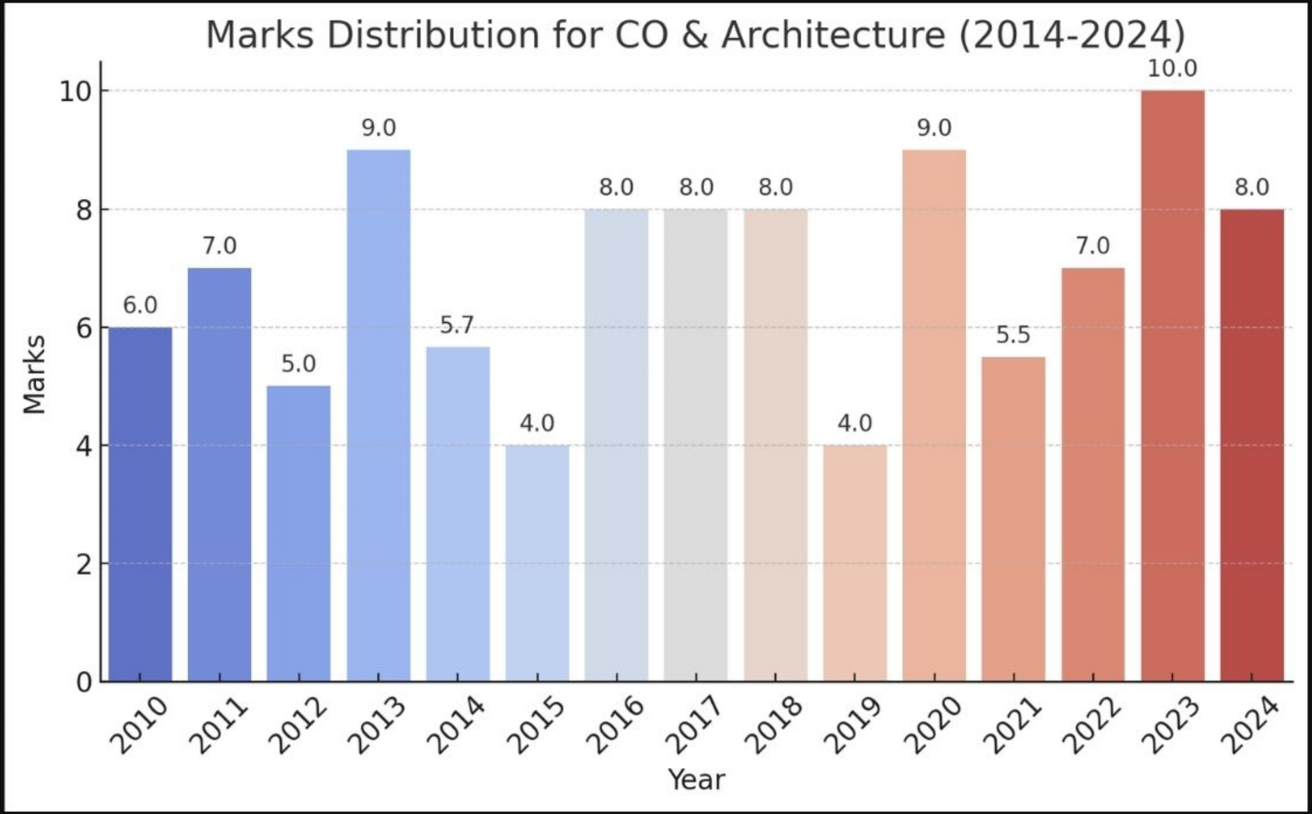


Important Topics

Compiler Design

- Grammar (50) easy
- Parsing (27) Tricky
- Syntax Sirected Translation (16) easy

Computer Organization and Architecture



Marks Distribution for CO & Architecture (2014-2024)
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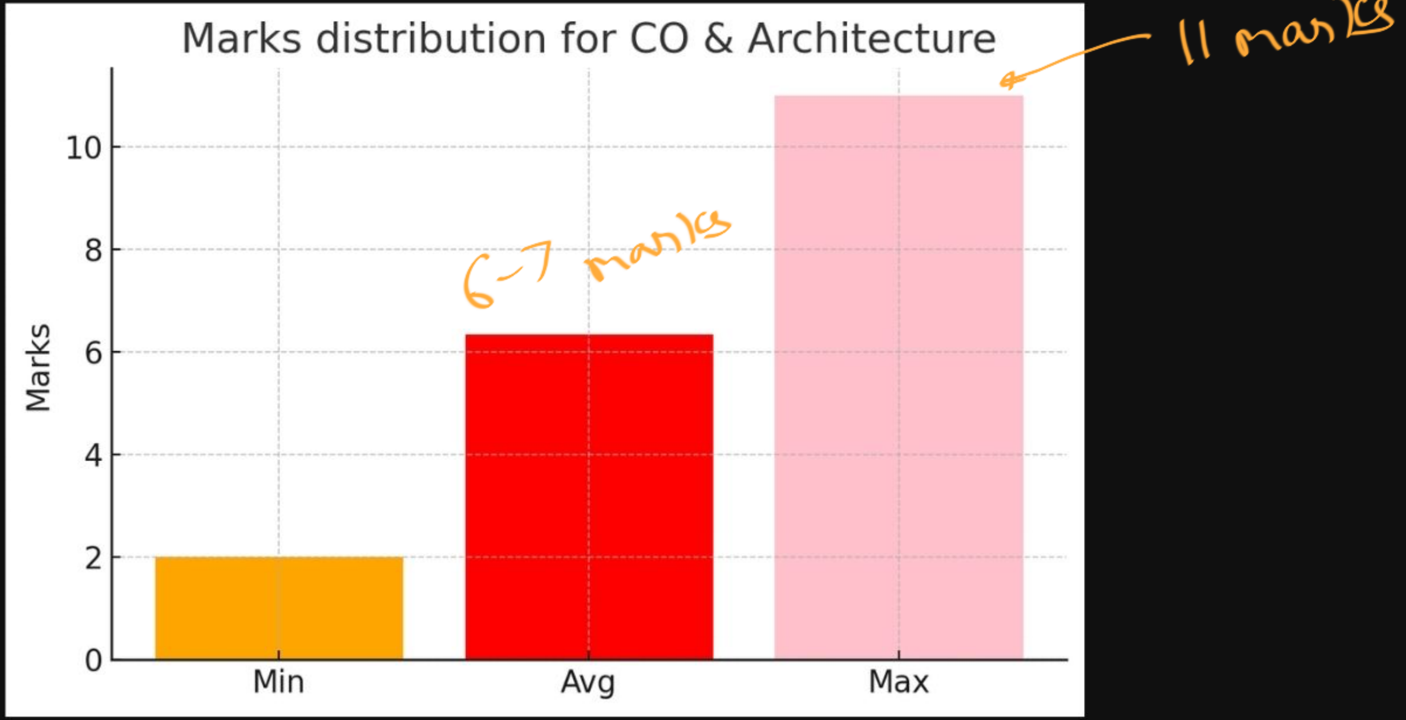
Computer Organization and Architecture

Min Max Avg marks of last 15 years



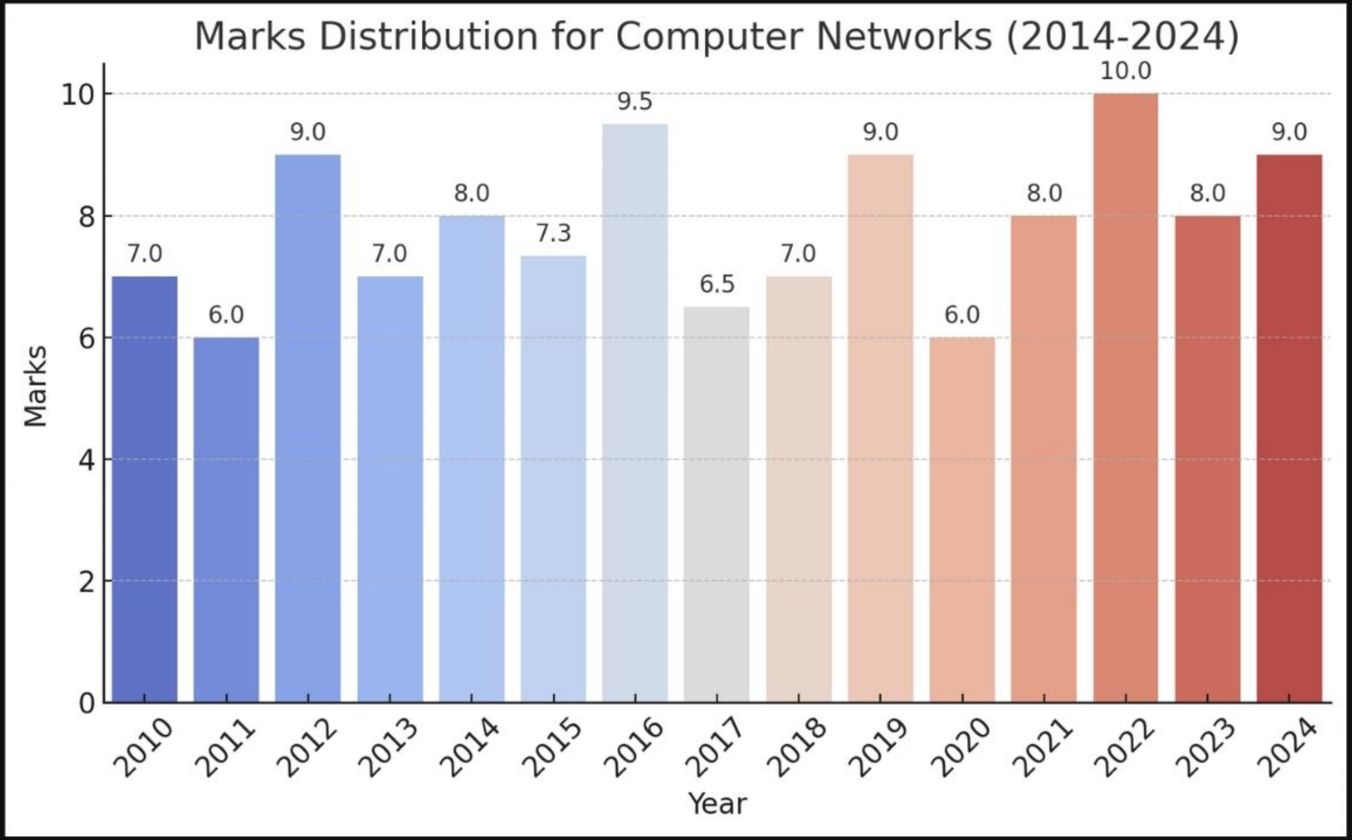
Important Topics

Computer Organization and Architecture

easy
- Addressing Modes (18)
- Cache Memory (71)
- Pipelining (42)
- Hazards

easy-mod

Computer Networks



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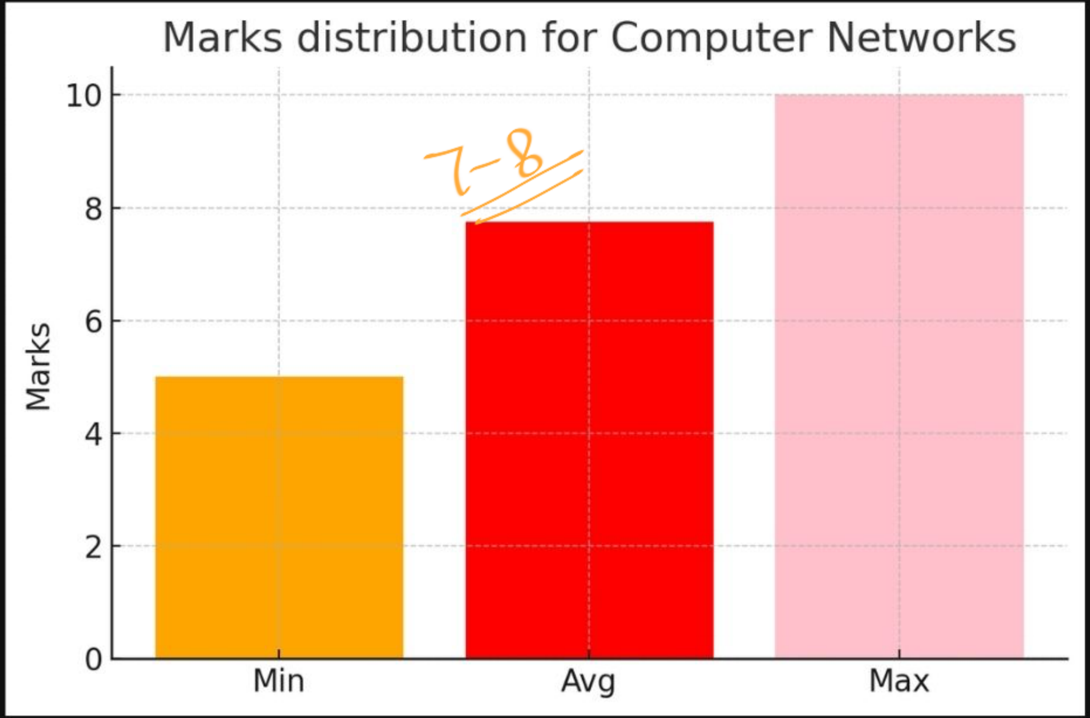
Computer Networks

Min Max Avg marks of last 15 years



Important Topics

Computer Networks

easy

- IP Addressing, Subnetting (30)

easy

- Application Layer Protocols (12)

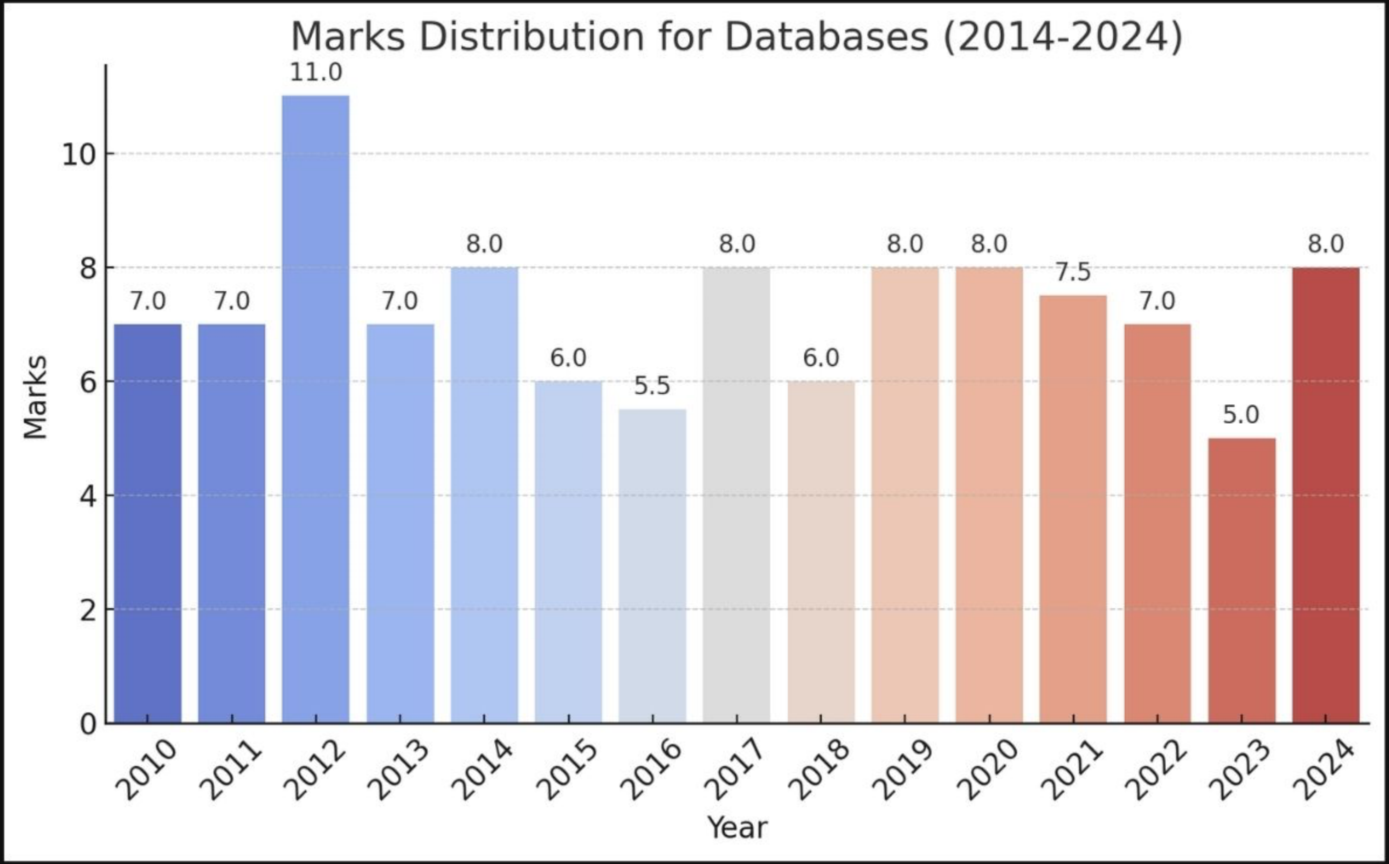
CRC → Tricky

- Error Detection (10)
- Routing Algorithms (Distance Vector, Link State) (21)

easy

- Sliding Window Protocols (21)
- TCP (28) → Congestion control

DBMS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DBMS

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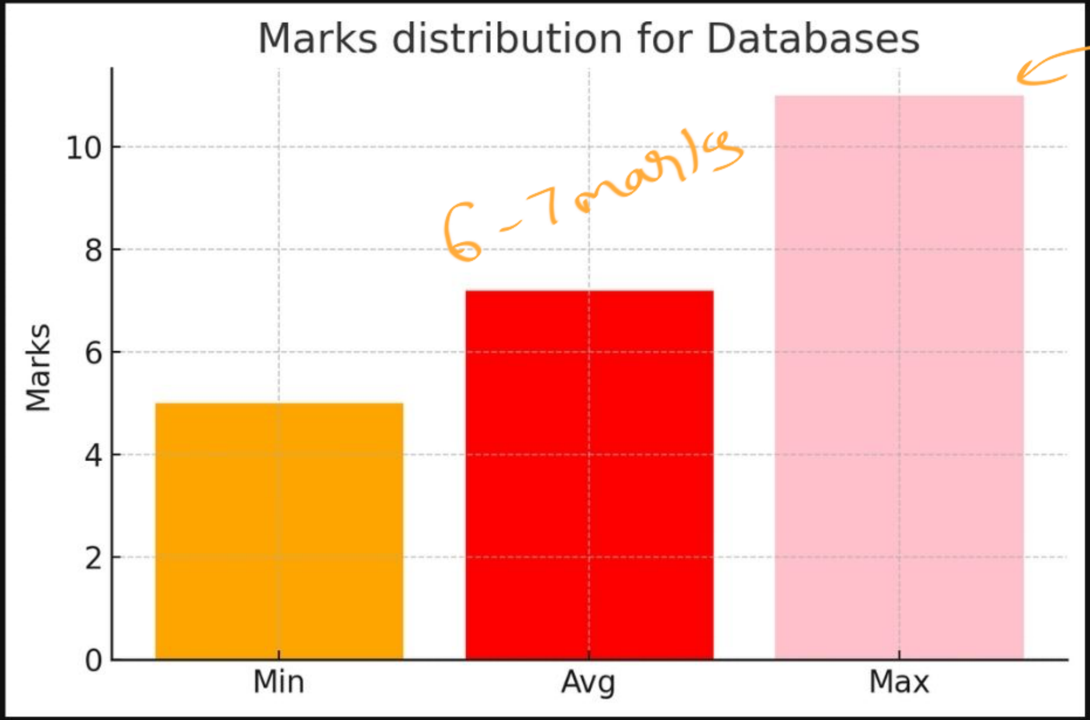


11 marks

Important Topics

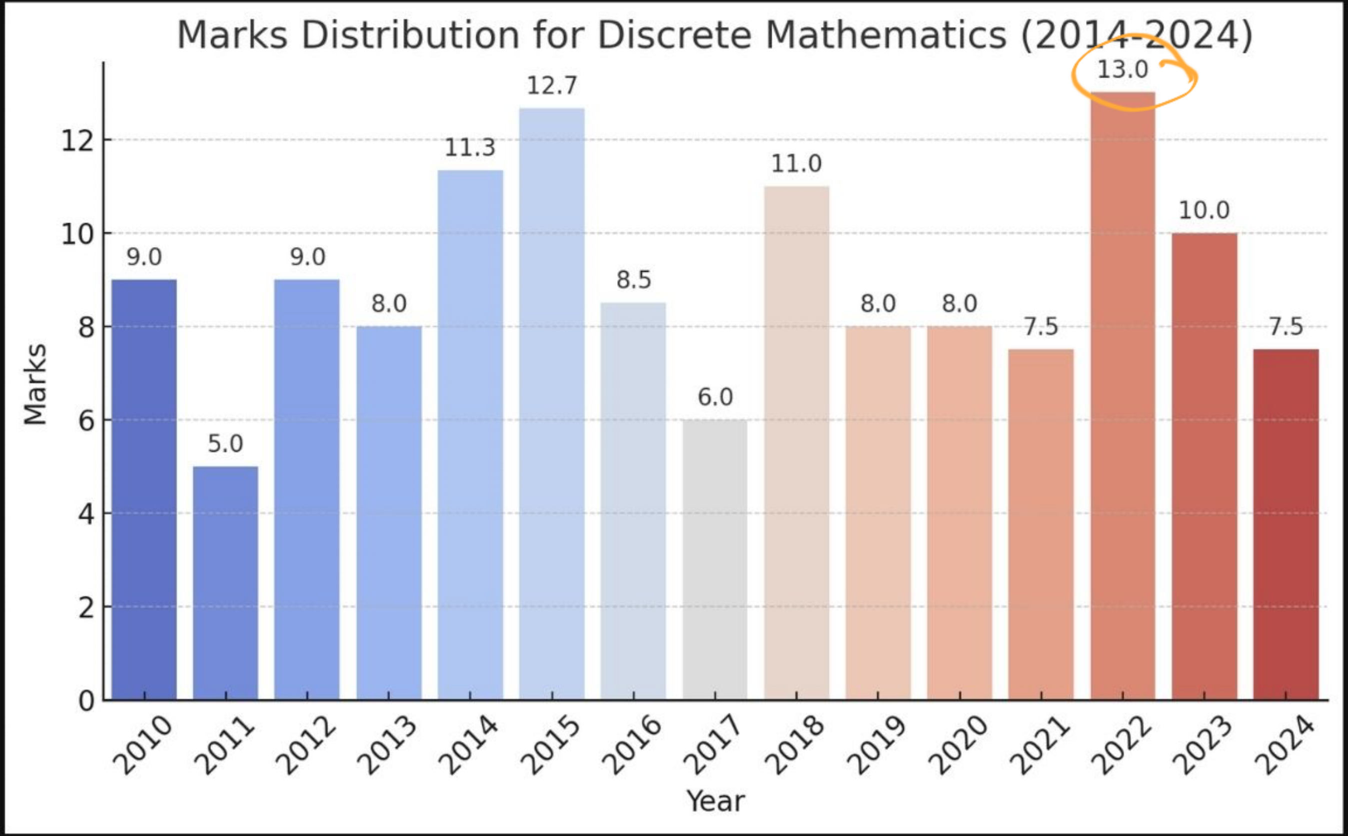
DBMS

Easy / Tricky

- Normalization (1NF, 2NF, 3NF, BCNF) (55)
- SQL Queries (55)
- Relational Algebra (29)
- B-Tree (29)
- Transactions and Concurrency Control (28)

easy

Discrete Mathematics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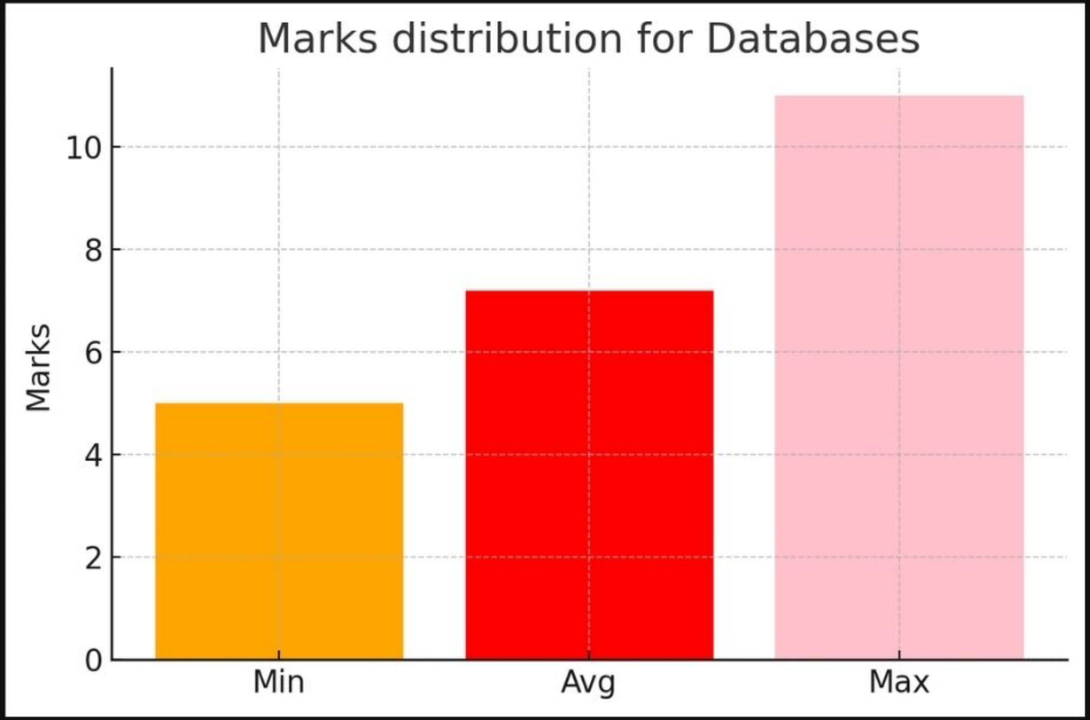
Discrete Mathematics

Min Max Avg marks of last 15 years

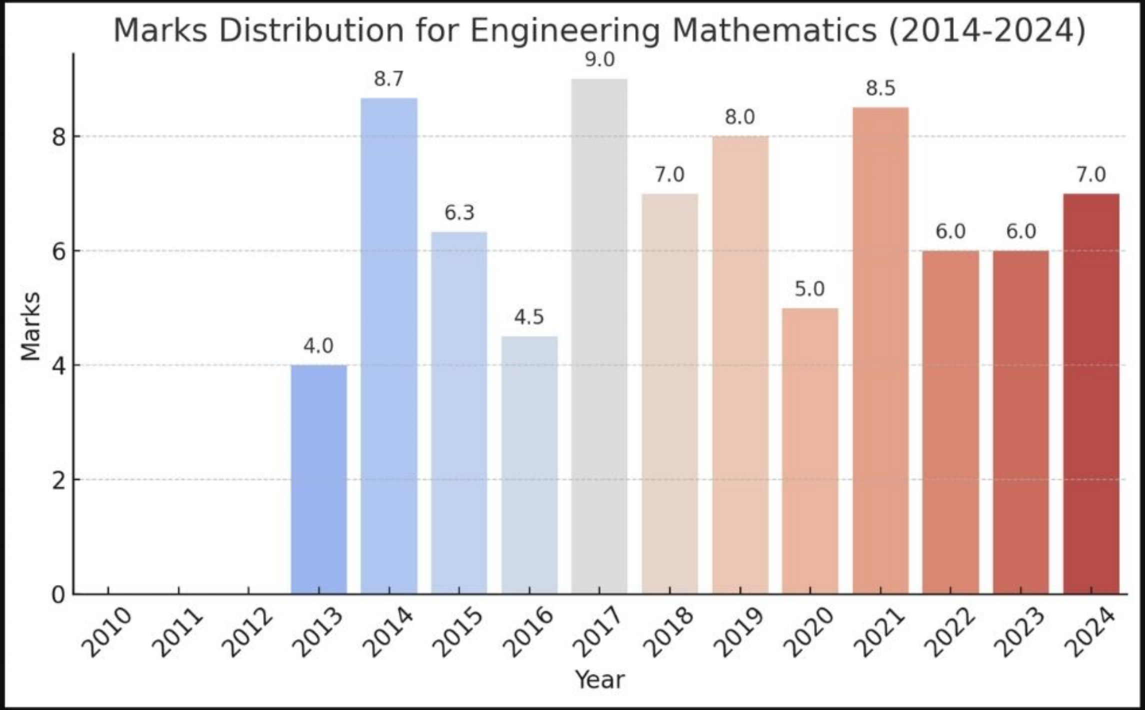


Important Topics

Discrete Mathematics

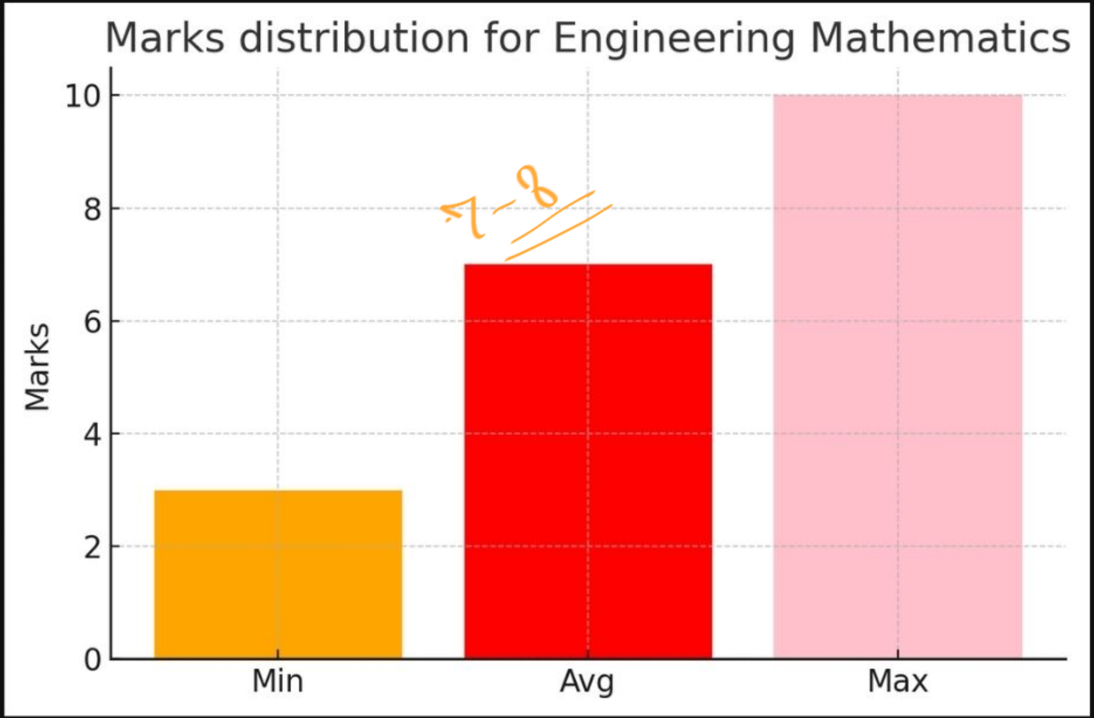
- Propositional Logic (40) ← easy
- First order Logic (32)
- Combinatorics (49)
- Set Theory (27) ← easy
- Graph Theory (81)
- Group Theory (33)
- Relations and Functions (65) ← easy

Engineering Mathematics


Marks Distribution for Engineering Mathematics (2014-2024)
Marks
Year
0
2
4
6
8
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
4.0
8.7
6.3
4.5
9.0
7.0
8.0
5.0
8.5
6.0
6.0
7.0

Engineering Mathematics

Min Max Avg marks of last 15 years



Important Topics

Engineering Mathematics

- Linear Algebra (Eigenvalues, System of Equations) (85)
- Probability (Bayes' Theorem, Conditional Probability) (88)
- Calculus (Limits, Continuity, Maxima Minima) (52)

Aptitude ✓

15 marks every year

fixed

Aptitude

- Solve Last 15 year PYQs of all the branches.

THANK YOU

Please let me know in the comments if you have any doubts related to GATE preparation.
Would be more than happy to answer them :)